

Internship Report – Day 10

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Internship Program: ParkNSecure – Smart Parking Solutions

Day 10 Overview:

Today, I trained an Automatic Number Plate Recognition (ANPR) model. The dataset included vehicle images with annotated license plates. The model learned to detect and extract number plates using object detection techniques. It will now be tested for accuracy and optimized for real-time use.

Tools Used Today

GEMINI, SONNET, Chatgpt, vs coder.

Task Given:

TO CREATE THE ANPR MODEL TO DETECT THE LETTERS AND NUMBERS PRESENT IN THE VEHICLES NUMBERPLATES.

Approaches:-

The ANPR system was approached using deep learning-based object detection models like YOLO or SSD. Preprocessing included resizing images and annotating number plate regions. The model was trained to detect and localize plates in diverse lighting and angle conditions. Post-processing involves OCR to extract text from the detected plate regions.

Attachments:-

train_anpr.py > ...

```
1 from ultralytics import YOLO
2 import os
3 import shutil
4
5 # Ensure model directory exists
6 os.makedirs("model", exist_ok=True)
7
8 # Minimal configuration for YOLOv8
9 CONFIG = {}
10 "model_type": "yolov8s",
11 "data_yaml": "Dataset/data.yaml",
12 "epochs": 25,
13 "batch_size": 8,
14 "img_size": 640,
15 "project": "model",
16 "name": "anpr_model",
17
18 # Main training function
19 def train_model():
20     print(f"Starting ANPR model training with {CONFIG['model_type']}...")
21
22     # Load a pre-trained YOLOv8 model
23     model = YOLO(f"{CONFIG['model_type']}.pt")
24     print(f"Model loaded: {CONFIG['model_type']}.pt")
25
26     # Clean previous training results if exist
27     if os.path.exists(os.path.join(CONFIG["project"], CONFIG["name"])):
28         print(f"Removing previous training output...")
29         shutil.rmtree(os.path.join(CONFIG["project"], CONFIG["name"]), ignore_errors=True)
30
31     # Start training with minimal parameters
32     print(f"Training on dataset: {CONFIG['data_yaml']}")
33     try:
34         results = model.train(
35             data=CONFIG["data_yaml"],
36             epochs=CONFIG["epochs"],
37             imgsz=CONFIG["img_size"],
38             batch=CONFIG["batch_size"],
39             project=CONFIG["project"],
40             name=CONFIG["name"],
41             exist_ok=True
42         )
43
44         print("Training completed")
45         model_path = os.path.join(CONFIG["project"], CONFIG["name"], "weights/best.pt")
46         print(f"Model saved at: {model_path}")
47
48         # Copy the model to an easy-to-access location
49         shutil.copy(model_path, "model/anpr_final.pt")
50         print("Model also saved to: model/anpr_final.pt")
51
52
```

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/25	0G	3.762	4.059	3.778	2	640: 100% ██████████ 386/386
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 37/37
	all	579	594	0.696	0.225	0.171 0.0645

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
2/25	0G	2.107	1.74	2.138	4	640: 100% ██████████ 386/386 [1:33:54<
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 37/37
	all	579	594	0.941	0.443	0.494 0.223

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
3/25	0G	1.753	1.416	1.691	1	640: 100% ██████████ 386/386 [1:03:05<
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 37/37
	all	579	594	0.966	0.455	0.546 0.272

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
4/25	0G	1.666	1.245	1.575	12	640: 59% ██████████ 228/386 [22:15<15]

3/25	0G	1.753	1.416	1.691	1	640: 100% ██████████ 386/386 [1:03:05<]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 37/37
	all	579	594	0.966	0.455	0.546 0.272
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
4/25	0G	1.645	1.187	1.551	4	640: 100% ██████████ 386/386 [52:09<00]
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% ██████████ 37/37
	all	579	594	0.659	0.604	0.617 0.31
Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
5/25	0G	1.569	1.108	1.49	11	640: 39% ███████ 149/386 [3:02:27<]

I have used the 4000 images dataset so it is taking too long to train now.